Free fall

antocciphysics | TEACHER KEY | CPA

Learning target: Describe and calculate ideal free fall near Earth.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 20; 4 course days later.*
$a=6$ m/s² right; $v_f=18$ m/s right. Constant net force and mass throughout.

02. Explain a worked example (worked)
Straight line of slope 10 m/s²; increasing speed means larger distances in successive equal times.

03. Look and explain (model)
Both accelerate at g and arrive together with equal initial conditions.

04. Build the solution (practice)
10, 20, 30 m/s; 10 m/s² downward.

05. Use the idea (reason)
Velocity is momentarily zero but gravitational acceleration remains downward.

06. Choose a method (mixed) *Review: Unit 1, Day 12; 12 course days later.*
Mass 2 kg; new acceleration 6 m/s². At fixed mass, acceleration scales with net force.

07. Exit check (exit)
40 m/s downward.

08. STRETCH: Apply it another way (stretch)
Both have the same increasingly spaced positions and acceleration; mass alone does not change ideal fall.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 20; 4 course days later.*
x interval [-44, -36] km; y interval [54, 66] km. Simplified geometry, gravity, Earth motion or missing observations limit the model. Bounds are not probabilities.

02. Explain a worked example (worked)
Straight line of slope 10 m/s²; increasing speed means larger distances in successive equal times.

03. Represent and interpret (model)
Both accelerate at g and arrive together with equal initial conditions.

04. Solve and justify (practice)
Top after 2 s; v=20-10(3)=-10 m/s.

05. Analyze the reasoning (reason)
Velocity is momentarily zero but gravitational acceleration remains downward.

06. Choose a method (mixed) *Review: Unit 1, Day 12; 12 course days later.*
Mass=1/slope≈0.5 kg; a=6 m/s². Resistance subtracts from driving force, producing an offset.

07. Exit check (exit)
25 m/s downward; displacement 5(2)+.5(10)(4)=30 m down.

08. STRETCH: Apply it another way (stretch)
Height 20+15(2)-5(4)=30 m. Solve 20+15 t-5 t²=0: t=4 s; negative root -1 s is not the flight time.